

22PH202 ELECTROMAGNETISM AND MODERN PHYSICS

UNIT III – LM 3.3 - GEOMETRICAL OPTICS

1.1 PRINCIPLE OF LEAST TIME

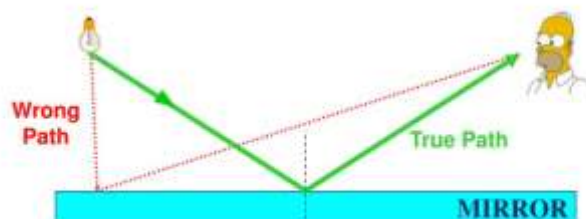
The principle of least time states that the light will travel from A to B in such a way that *the time is minimized*. Using this principle you can derive a whole bunch of optics laws. But to me that doesn't make any sense.

1. Fermat's principle works for reflections
2. So let's solve it for the case when light enters another medium
3. Yay we just derived Snell's law!

Problem is that they never derived Fermat's principle for anything else than reflection. And it hardly works for reflections: why doesn't the beam just travel straight from A to B without reflecting in the mirror

Fermat's Principle of Least Time

Light travels in **straight lines** and will take the **quickest** path to get from point A to point B



1.2 GEOMETRICAL OPTICS

Geometrical optics is the treatment of the passage of light through lenses, prisms, etc. by representing the light as rays. A light ray from a source goes in a straight line through the air, but when it encounters a lens, prism, or mirror it bends or changes direction.

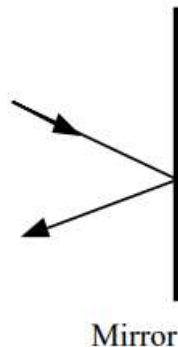


Figure-1 Geometrical optics

There are formulas and rules that tell how much the change in direction will be for any circumstance of a ray passing into or out of a medium, lens, prism, etc. With these rules and a little geometry, one can construct the whole path of a ray, hence the name Geometrical Optics.

1.3 IMAGES PRODUCED BY LENSES CAN BE ONE OF TWO TYPES:

- A real image
- A virtual image

1.3.1 Real Images

- A real image is defined as:

An image that is formed when the light rays from an object converge and meet each other and can be projected onto a screen

- A real image is one produced by the convergence of light towards a focus
- Real images are always inverted
- Real images can be projected onto pieces of paper or screens.
- Real images are where two **solid lines** cross in ray diagrams

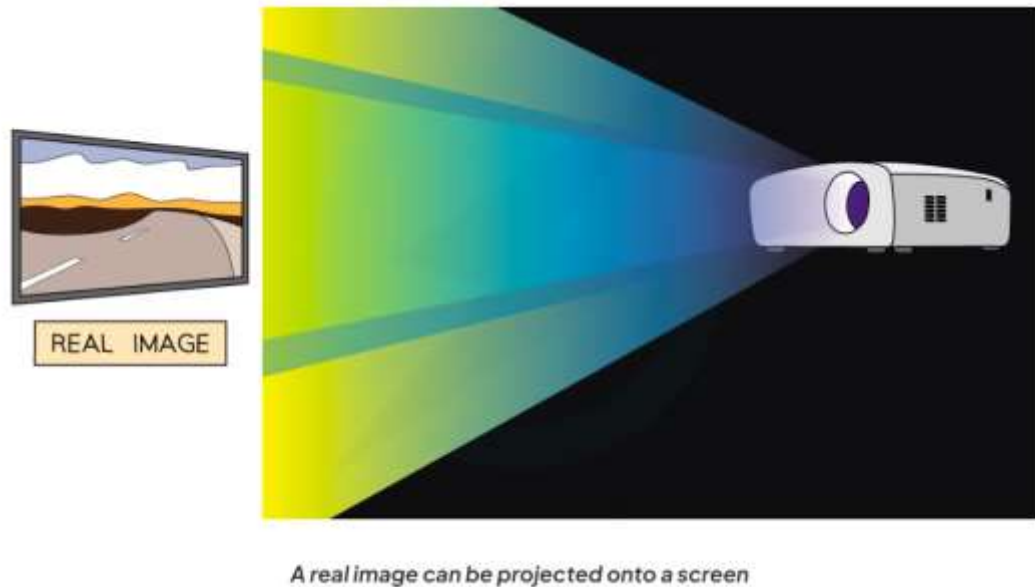


Figure-2 An example of a real image is the image formed on a cinema screen

1.3.2 Virtual Images

- A virtual image is defined as:

An image that is formed when the light rays from an object do not meet but appear to meet behind the lens and cannot be projected onto a screen

- A virtual image is formed by the divergence of light away from a point
- Virtual images are always upright
- Virtual images cannot be projected onto a piece of paper or a screen.
- Virtual images are where two **dashed lines**, or **one dashed and one solid line** crosses in ray diagrams



A reflection in a mirror is an example of a virtual image

Figure-3 An example of a virtual image is a person's reflection in a mirror

1.4 CONVEX & CONCAVE RAY DIAGRAMS

- Lenses can be used to form images of objects placed in front of them
- The location (and nature) of the image can be found by drawing a ray diagram:

1.4. 1 Convex Lens Ray Diagrams

- If an object is placed **further** from the lens than the focal length f then a **real** image will be formed, and the converging lens ray diagram will be drawn in the following way:

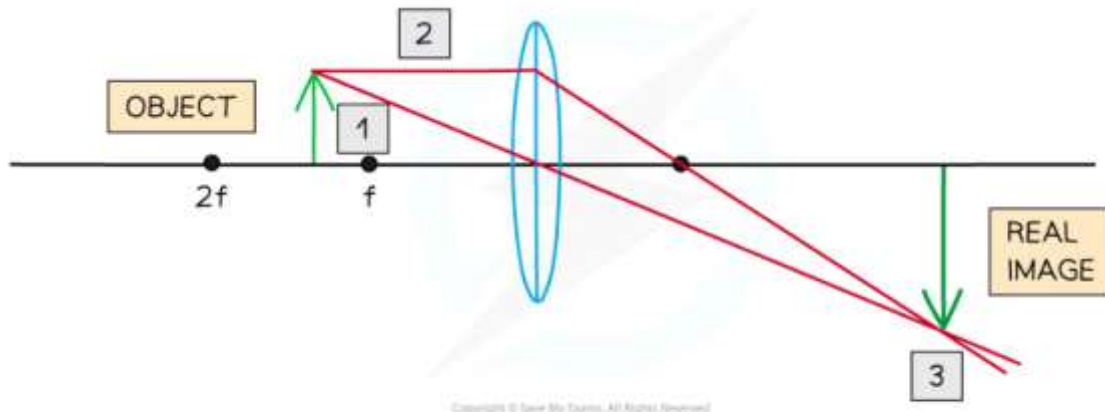
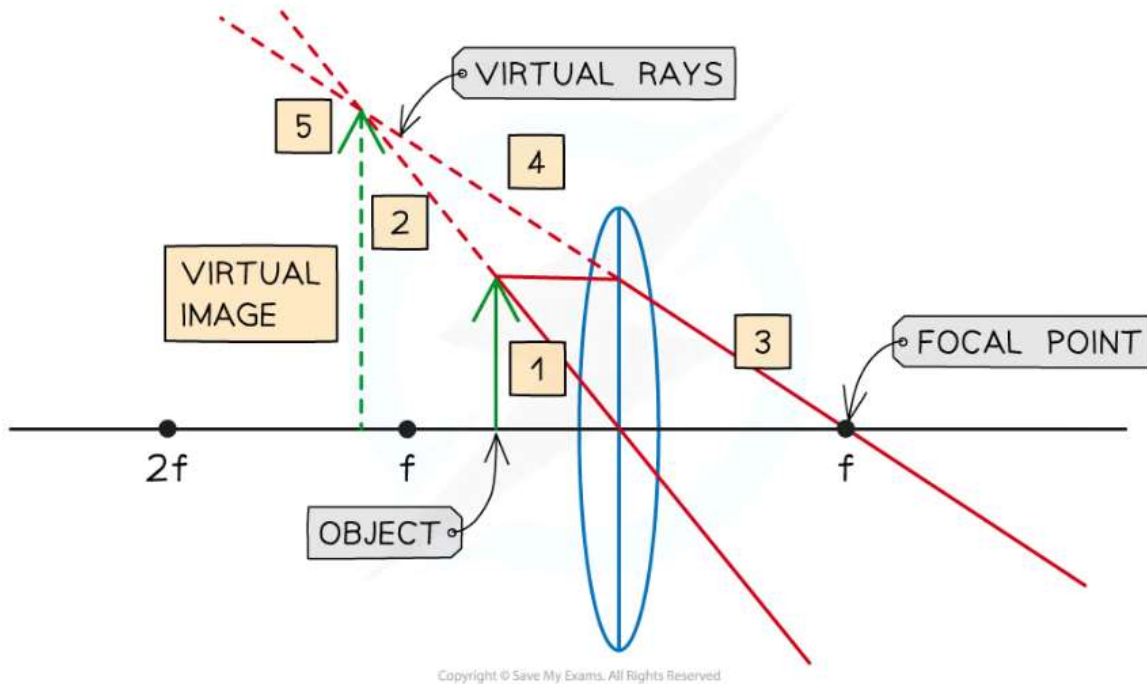


Diagram showing the formation of a real image by a lens

Figure-4 Convex Lens Ray Diagrams

start by drawing a ray going from the top of the object through the **centre** of the lens. This ray will continue to travel in a straight line

1. Next draw a ray going from the top of the object, travelling parallel to the axis to the lens. When this ray emerges from the lens it will travel directly through the **principal focus** f
 2. The image is the line drawn from the axis to the point where the above two rays **meet**
- When describing an image, consider if it is:
 - **Real** or **virtual**
 - **Magnified** (larger) or **diminished** (smaller)
 - **Upright** or **inverted**
 - For example, in the above diagram, the image is:
 - **Real**: the light rays meet each other after refraction
 - **Magnified**: the image is larger than the object
 - **Inverted**: the image is formed on the opposite side of the principal axis
 - A convex lens will produce a **real** image of an object which is placed at a distance **greater** than the **focal length** from the lens
 - If the object is placed **closer** to the lens than the focal length f then a **virtual** image will be formed and the converging lens ray diagram will be drawn in the following way:



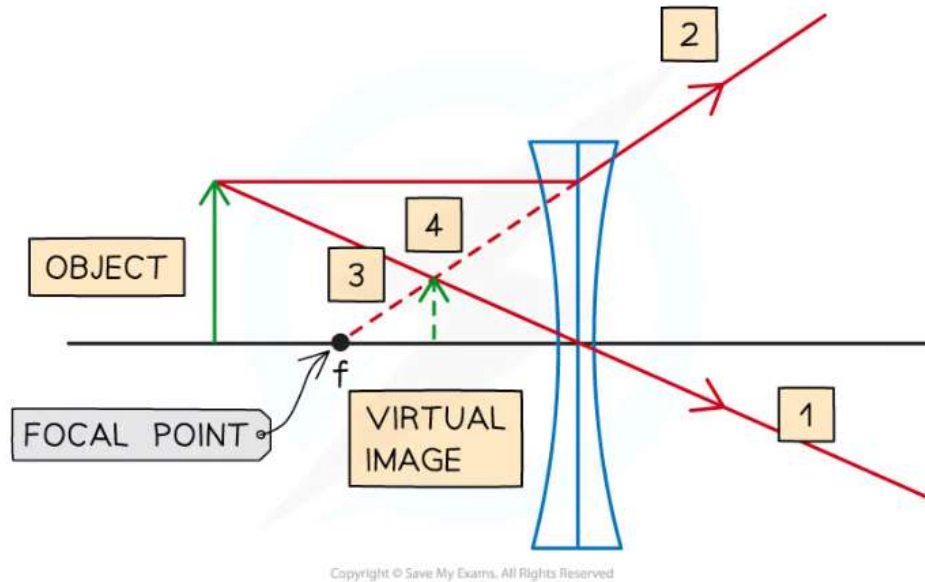
A virtual image formed when the object is placed closer than the focal length

1. Start by drawing a ray going from the top of the object through the **centre** of the lens. This ray will continue to travel in a straight line
 2. Draw a dashed line continuing this ray upwards
 3. Next draw a ray going from the top of the object, travelling parallel to the axis to the lens. When this ray emerges from the lens it will travel directly through the **principal focus** f
 4. Also, draw a dashed line continuing this ray upwards
 5. The image is the line drawn from the axis to the point where the two dashed lines **meet**
- In this case, the image is:
 - **Virtual:** the light rays appear to meet when produced backwards
 - **Magnified:** the image is larger than the object
 - **Upright:** the image is formed on the same side of the principal axis

1.4. 2 Concave Lens Ray Diagrams

- Concave (diverging) lenses can also be used to form images, although the images are always **virtual** in this case

- If an object is placed **further** from the lens than the focal length f then a concave lens ray diagram will be drawn in the following way:



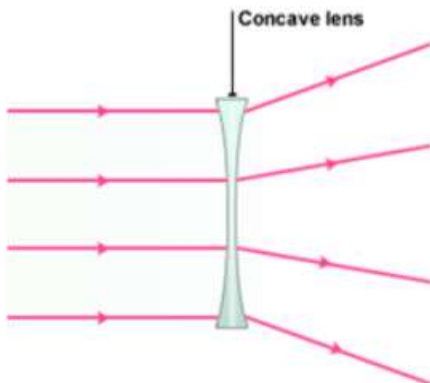
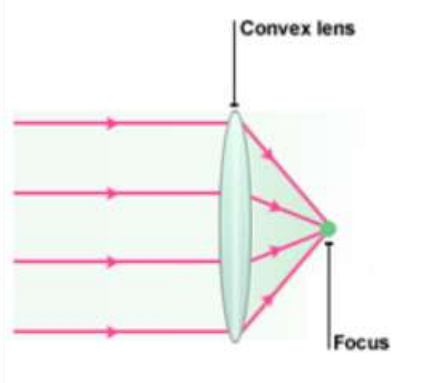
Concave lenses only produce virtual images

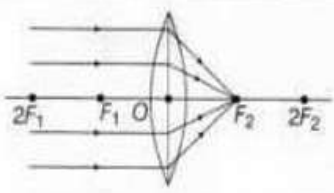
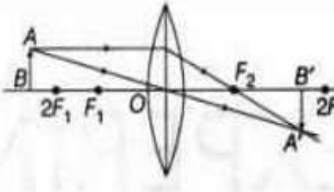
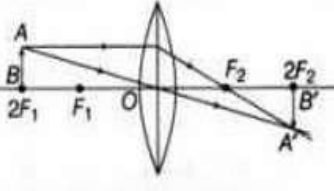
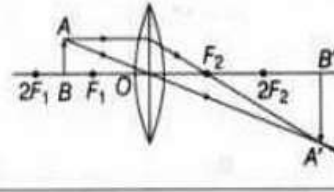
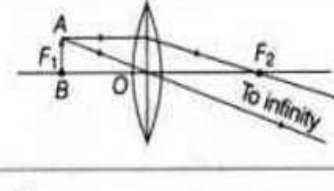
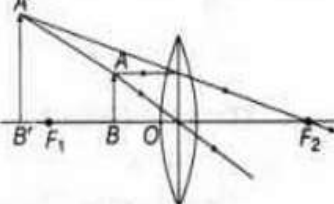
Figure-5 Concave Lens Ray Diagrams

- Start by drawing a ray going from the top of the object through the **centre** of the lens. This ray will continue to travel in a straight line
 - Next draw a ray going from the top of the object, travelling parallel to the axis to the lens. When this ray emerges from the lens it will travel directly upwards away from the axis
 - Draw a dashed line continuing this ray downwards to the focal point, f
 - The image is the line drawn from the axis to the point where the above two rays **meet**
- In this case, the image is:
 - Virtual:** the light rays appear to meet when produced backwards
 - Diminished:** the image is smaller than the object
 - Upright:** the image is formed on the same side of the principal axis

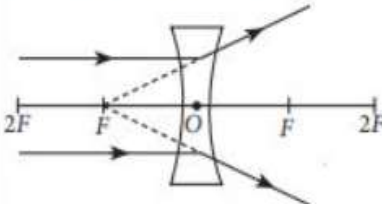
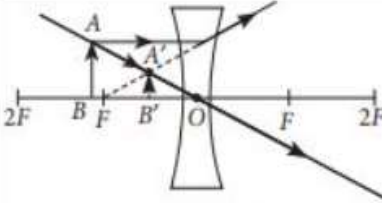
1.5 COMPARING CONVEX & CONCAVE LENSES

- The image produced by a **convex** lens can be either **real** or **virtual**
 - This means the image can be inverted (real) or upright (virtual)
- The image produced by a **concave** lens is always **virtual**
 - This means the image will always be upright

Concave Lens	Convex Lens
1. It is the type of lens that disperses the light rays around, hitting the lenses.	1. It is the type of lens that merges the light rays at a particular point, traveling through it.
2. This lense is used to correct myopia	2. This lens is used to correct hypermetropia
3. It has a negative focal length	3. It has a positive focal length
4. It forms a virtual, erect and magnified image	4. It forms a real and inverted image.
5. This lens makes objects appear smaller and farther	5. This lens makes an object appear closer and larger
6. 	

Position of object	Position of image	Size of image	Nature	Ray diagram
At infinity	At F_2	Extremely diminished	Real and inverted	
Beyond $2F_1$ (at finite distance)	Between F_2 and $2F_2$	Diminished	Real and inverted	
At $2F_1$	At $2F_2$	Same size	Real and inverted	
Between F_1 and $2F_1$	Beyond $2F_2$	Magnified	Real and inverted	
At F_1	At infinity	Highly magnified	Real and inverted	
Between lens and F_1	On same side of object	Magnified	Virtual and erect	

Concave lens

	Ray diagram	Position of object	Position of image	Nature of image
(a)	 $u = -ve, v = -ve$ and $f = -ve$	At infinity	At F	Virtual, erect and highly diminished
(b)	 $u = -ve, v = -ve$ and $f = -ve$	Between infinity and O	Between F and O	Virtual, erect and diminished